COMPX551 Assignment 4

Muse Headset

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How a frame is written

The argument to `frame` is the frame title. Body text is plain LaTeX: a blank line starts a new paragraph, and metropolis spaces them for you.

- A bullet list is the usual `itemize`
- alert colours a phrase in the theme's accent

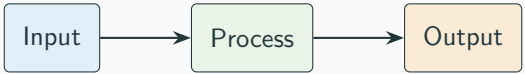
A fixed-width block, for a command or a config line:

```
docker run --rm -it alpine sh
```

Frame options worth knowing

- `\begin{frame}[fragile]` is required when the body holds a `verbatim` or `lstlisting` environment that beamer cannot read ahead over
- `[plain]` drops the frame title, footer and progress bar, for a full-bleed figure
- `[standout]` inverts the frame, for a section break or a closing slide
- `[allowframebreaks]` splits an overlong frame across several slides

A figure scaled to whatever space is left:



Application and its Purpose

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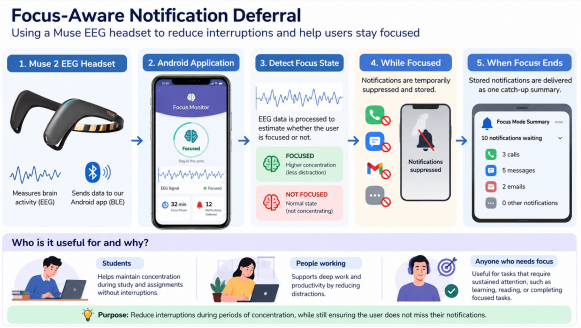
Chosen Device

Muse 2 EEG Headset

Provides EEG data that our Android application uses to estimate the user's focus state.

What does our application do?

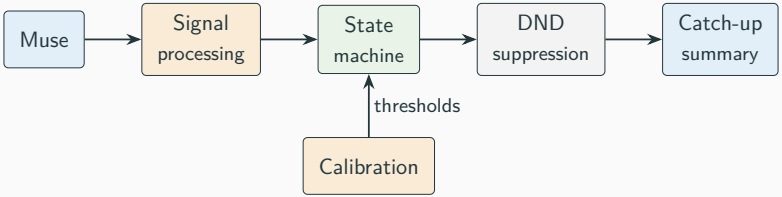
- Estimates whether the user is focused using EEG data.
- Temporarily suppresses incoming notifications while focused.
- Stores notifications during the focused period.



Architecture and device integration

System overview

EEG in, a focus state out, one policy applied consistently downstream.



- Runs entirely on-device, with no data leaves the phone
- One state drives both suppression and the catch up summary
- Fail open at every stage, a bad signal never silently holds a notification

Connecting to the headset

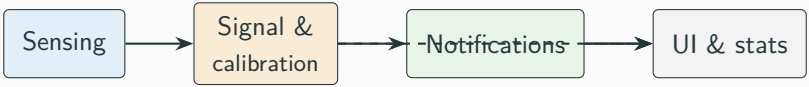
A standalone BLE peripheral, not a paired Wear OS watch, so we use Muse's SDK (`LibMuse`) rather than the Wear OS Data Layer API.



- Registers only the packet types needed, nothing Muse specific leaks out
- Small connection state machine, rescans automatically on a drop
- A dropped or poor contact signal reads as unknown, never silently nothing

Built as four independent modules

The pipeline's own stages became the team's module boundaries — one interface per stage, one owner, developed and tested in isolation.



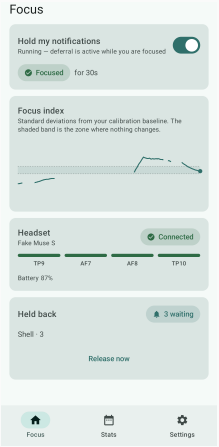
- Every interface has access to a fake demo, so all four people started building day one
- An OSC bridge to an existing app gave real EEG data before BLE worked
- Swapping any module is a one line change, nothing downstream changes

Data processing

Data processing

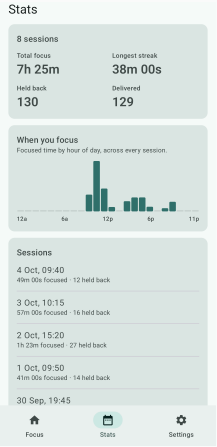
Visualisation and summary data

Live view: what the headset is saying right now



- Focus index: the last two minutes at 1 Hz, in standard deviations from the user's own calibration baseline
- The shaded band is the hysteresis zone between the two thresholds, where nothing changes
- Artifact-rejected epochs *break* the line rather than being interpolated over; the y-range is fixed at ± 3 SD so a calm signal looks calm
- State pill with time-in-state, per-electrode contact quality, and the held-back queue as "app · count", never message content

Summary data: what a week of focus looks like



- One record per session: focused time, longest streak, notifications held back and delivered
- Focused time is measured between state transitions, not by counting FOCUSED epochs, so a dropped packet never shortens the total
- When you focus: focused minutes by hour of day, summed across every session, so the user can see when they do their best work
- Stored on-device as one JSON line per session, written once at session end; every total is computed on read, nothing leaves the phone

Cultural and ethical considerations

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Reflection and future work

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Thanks